

System-Level Design Optimization of a Hybrid Tug

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Abstract—Designing a new vessel is a complex multi-objective design process. It involves knowledge from different fields, like naval architecture and mechanical engineering. Assessment of an optimal design for more complex topologies than a conventional Diesel powertrain becomes more difficult due to the increased number of powertrain components and feasible combinations. The purpose of this work is to present a system-level design methodology, which speeds up the component sizing and control for a powertrain topology. This system-level design problem is formulated into the minimization of a cost function. The cost function consists of the costs of the different powertrain components together with the operational costs over a specified operational profile. For the sizing of the battery and control parameters, the use of a convex optimization algorithm ensures a global optimum is found very quickly in the search space at relatively low-computational effort. For the ON/OFF switching of the Diesel engine and the Diesel generator set, an iterative scheme using convex and mixed-integer optimization is proposed. Here, without loss of generality, the optimization case study is presented for a hybrid tug.

Index Terms—Optimization, Convex modeling, Mixed-integer problem, Hybrid Electric Ships, Powertrains, Optimal Control, Dynamic Programming

I. INTRODUCTION

Designing a new vessel is a complex multi-objective design process. It involves the knowledge from different fields, like naval architecture and mechanical engineering [1]. Since, vessels do not have prototypes, the first is expected to be already profitable for its purpose [2]. The traditional design process of a new vessel is often depicted with a design spiral [3]. The design spiral consists of iterations through several design tasks. With each iteration the vessel design is refined towards the next stage in the design process [4]. During the first stage, the initial concept design is formulated mainly by experience and basic calculations [5]. However, a hybrid powertrain, as a more complex powertrain topology, increases the amount of feasible powertrain topologies significantly. A methodology to evaluate each feasible topology for optimality within a reasonable amount of time would enable the ship manufacturer to evaluate more feasible powertrain topologies.

A. System-level design methodology

System-level design (SLD) for hybrid powertrain topologies can be divided into three different coordination architectures: (i) alternating; (ii) nested; and, (iii) simultaneous [6]. In the alternating architecture, first, the plant is optimally designed and for this design, the controller is optimally tuned, accordingly. This is a process, which is repeated iteratively.

Within the nested architecture, for every evaluation of a plant, the controller is fully optimized (assuming a unidirectional coupling); and, for the simultaneous architecture, the controller design and component sizing is done simultaneously.

In this work, the development of a mathematical model for calculating the optimal sizing and control parameters of a new vessel for a fixed powertrain topology is discussed. For this case study, the simultaneous architecture, is proposed, where this SLD problem is formulated with a cost minimization function. This cost function minimizes both the component and operational costs. In a similar manner as in [7], [8], [9], the use of a convex optimization algorithm ensures a global optimum is found very fast in the search space at relatively low-computational effort. However, in this paper, the inherently nonconvex control problem, due to present engine ON/OFF switching state, is iteratively solved using convex and mixed-integer (MI) solvers. Here, without loss of generality, the optimization case study is a hybrid tug (Cf. Fig. 1) that is designed by Damen Shipyards (Gorinchem, The Netherlands). As a benchmark, a conventional Diesel powertrain topology is used. Before discussing the modeling and the approach, first, the powertrain functionality is described below.

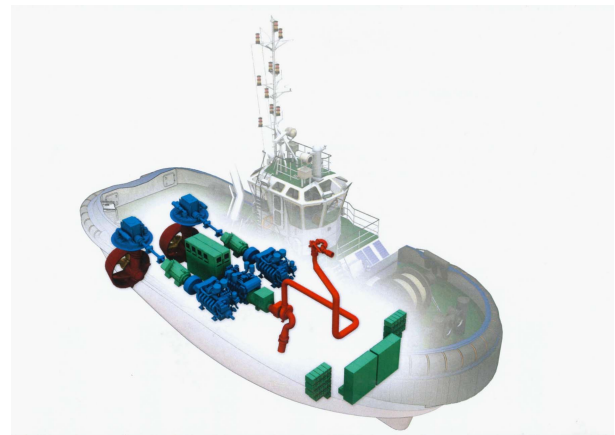


Fig. 1: Hybrid tug (Courtesy of Damen Shipyards).

B. Hybrid powertrain topology

The powertrain of the vessel [10] is shown in Fig. 2. The powertrain consists of one Diesel engine (DE, 690-kW) - generator (G, 650-kWe) set, denoted as GE set, supplying the hotel (H) or auxiliary loads (AUX), charging the battery

pack (BAT, 240-kWh) and powering the electric induction machines (EIM, 230-kWe) via a main switch board (MSB, 400-V, 60-Hz). The MSB is connected via power converters to two EIMs that are mechanically connected to the propeller shafts for motor assisting/charging in parallel with the two other remaining Diesel engines (DE, 1600-kW each). Clearly, due to redundancy requirements, comfort and other technical requirements, the topology is typically different from the layout of vehicle powertrain systems (equipped with a single engine). For simplicity, the total load at the propeller side is assumed to be equally distributed over both propellers resulting in a reduced model for optimization. The vessel can

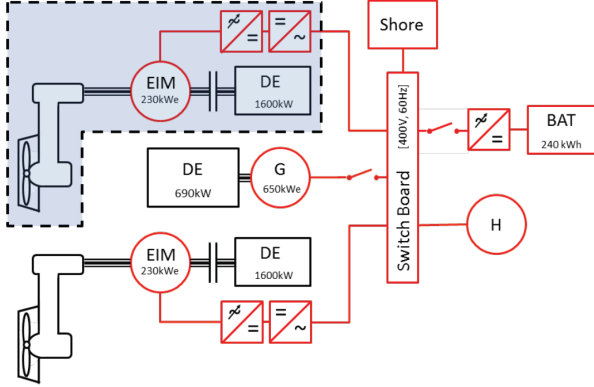


Fig. 2: Powertrain block model of the hybrid tug [10].

be operated in five distinct hybrid operation modes. The GE set enables the powertrain to operate in (i) Diesel-electric propulsion mode (EP_{de}) [11] and (ii) GE charging mode (CH_{ge}). During EP_{de} mode, the 1600-kW DE's are assumed to be OFF and decoupled (via clutches) from the drive shafts (eliminating engine drag torques). In the Diesel-electric mode, the GE is directly powering (as in a series hybrid) the electric (induction) motors (EM); and, in the CH_{ge} mode, the GE is charging the BAT as well (DE can be ON or OFF and decoupled). Other (conventional or parallel hybrid) modes are: (iii) DE-only mode (DE) (GE is switched OFF), (iv) DE-charging mode (CH_{de}) (GE can be switched ON or OFF), and (v) electric machine pure-electric mode (EP_{em}) (DE and GE are both OFF and decoupled). Note that the propeller is not used for regenerating energy via 'braking'. A propeller can regenerate energy via 'wind milling', however, the potential amount of energy harvested via wind milling, while slowing down is negligible small [12]. The optimal control problem focusses on minimization of the total fuel power, P_f and is determined by the DE and GE ON/OFF states, $\theta_i \in \{1, 0\}$ and the battery control input power P_b ,

$$P_f = \theta_{de} \cdot P_{f,de} + \theta_{ge} \cdot P_{f,ge} = \frac{P_p}{\eta_{de} \cdot \eta_t} - \frac{P_{em}}{\eta_{de}} + \frac{P_{ge}}{\eta_{ge}}, \quad (1)$$

with the electric machine powers, P_{em} during charging (+) or discharging (-):

$$P_{em} \in \{P_{em}^+, P_{em}^-\} = \left\{ \frac{P_b^+ + P_{ge}}{\eta_{em}^+}, (P_b^- + P_{ge}) \eta_{em}^- \right\}, \quad (2)$$

if the auxiliary power P_{aux} at the input of the electric machine(s) for simplicity is set to zero. Moreover, the battery could also be recharged via the grid with charging power P_g at the shore at operation off-duty hours (Cf. Fig. 2). The (given) propulsion power $P_p \in \mathbb{R}^+$, as exogenous input, strongly depends on the ship (hull), mass, velocity, propeller design, and other vessel characteristics. In this paper, we provide also a convex model for describing the drive shaft torque demands at a given vessel velocity for a particular propeller design. The rest of the paper focusses on the development of convex optimization models and system-level design for hybrid powertrain optimization of vessels. First, the system modeling and operational power demand will be discussed in Section II. The convex models together with the optimization scheme and algorithms developed for solving the integrated powertrain sizing and control design problem are described in Section III. The results are presented for determination of the optimal battery sizes, yet without loss of generality can be extended to other component sizings. Finally, in Section IV, the conclusions are discussed.

II. SYSTEM MODELING AND OPERATIONAL DEMAND

Compared to vehicles, it is not only the powertrain layout, which is different, yet, additionally, the load demand is mainly determined by the vessel speed v_s and the bollard pull over time. The bollard pull (BP) is a conventional measure of the pulling (or towing) power of a tug. These two inputs create an operational profile.

A. Vessel model

When the vessel is sailing, it experiences a resistive force from the water acting on the hull. The resistive force F_r consists of three components: the frictional resistance, form resistance and the wave resistance [13]. For the hybrid tug, a resistance curve derived from measurements is available. This curve contains the frictional, form and wave resistance versus the vessel speed, $F_r(v_s)$. This wave resistance is the resistance caused by waves created by the vessel itself. The resistance from external waves is not included and highly unpredictable. In bollard pull conditions, a towing force is added to the overall vessel force balance, i.e.,

$$F_p \cdot (1 - c_{td}) \cdot N - F_r(v_s) - F_{bp} = m_v \cdot a_v, \quad (3)$$

with the thrust force F_p (N), the bollard pull F_{bp} (N), the resistance force $F_r(v_s)$ (N), the term c_{td} being the thrust deduction factor to compensate for thrust deduction (caused by suction effect of the propeller increasing the hull resistance), the number of propellers N (-), the mass of the vessel m_v (kg), and the vessel velocity change a_v (m/s²).

B. Propeller model

Using the propeller diameter D (m), type and calculated thrust force at the propeller(s) F_p (N), the parameters to calculate the required propeller torque T_p (Nm) and rotational speed n_p (rpm) can be found. The parameters are related to the

non-dimensional thrust-coefficient K_F and torque-coefficient K_T , respectively:

$$K_F = \frac{F_p}{\rho \cdot n_p^2 \cdot D^4} \quad ; \quad K_T = \frac{T_p}{\rho \cdot n_p^2 \cdot D^5} \quad (4)$$

with the thrust T_p (N) and the density of water ρ (kg/m³). The coefficients do not only vary with n_p and D , however, also with the advance ratio J (-),

$$J = \frac{v_a}{n_p \cdot D}, \quad (5)$$

with the advance velocity v_a (m/s) derived from v_s with the (constant) wake factor¹ c_w (-) [13],

$$v_a = (1 - c_w) \cdot v_s \quad (6)$$

The advance velocity is the speed of the vessel relative to the water mass around it, lower than the observed speed of the vessel. The wake factor is the difference of speed between v_s and v_a expressed as a percentage of v_s , related to the hull shape and number of propellers. Using these data, second order polynomials can be fitted resulting in empirical functions f, g that well captures the relationship between $f : J \mapsto K_F$ and $g : J \mapsto K_T$. The thrust force (N) becomes,

$$F_p = \rho \cdot n_p^2 \cdot D^4 \cdot f(J(\cdot)) \approx \rho \cdot n_p^2 \cdot D^4 \cdot \sum_{i=0}^{i=2} a_i \cdot J^i(\cdot) \quad (7)$$

It can be seen that for a certain thrust force F_p and advance velocity v_a , the propeller rotational speed n_p can be calculated as a positive root of Eq. (7). Given the inputs n_p and v_a , accordingly, the value for T_p can be found in a similar fashion. Note that T_p and n_p can be found for different propeller designs by simply changing the propeller specifications K_F, K_T and D . For reasons of confidentiality the vessel parameters, characteristics ($D, m_v, c_{td}, F_r, c_w, P_{aux}$), and the fitted coefficients a_i will not be provided here.

C. Exogenous model inputs

Taking into account the gear ratio, r_g and transmission efficiency of the azimuth thruster, η_t , the shaft torque $T_s(t) = T_p(t)/(r_g \cdot \eta_t)$ and rotational speed $n_s(t) = n_p(t) \cdot r_g$ over time t are derived from the previously calculated $T_p(t)$ and $n_p(t)$ at the propeller side. This results in a series of data sets,

$$\Lambda(t) = [T_s(t) \ n_s(t)] \text{ for } t \in [t_0, t_f], \quad (8)$$

with $[t_0, t_f]$ representing the length of the operational profile. The time-dependent data set $\Lambda(t)$ is an (exogenous) input for the powertrain optimization that is assumed to be known.

¹The wake factor has a positive effect on the hull resistance by effectively lowering the water velocity near the propeller intake creating a larger impulse to the water.

III. CONVEX MODELING AND OPTIMIZATION

The optimization layer consists of three levels. Within the level (a), as depicted in Fig. 3, the sizing of the DE, s_{de} (-) and GE, s_{ge} (-), is here determined via an exhaustive search method. Exhaustive search can be more computational expensive, however, scaling the available GE sets and DEs was found to be unacceptable (inaccurate) due to the strongly varying component characteristics for the hybrid tug. Therefore, we will assume a (relative) constant size for both components, i.e., $s_{de} = s_{ge} = 1$ (-). The EM and BAT are scalable and can be modelled by a convex function [14] sufficiently accurate. Accordingly, the scaling of the EM and BAT can be optimized simultaneously with the control variables regarding the torque and power split. This takes place in the level (b). The ON/OFF switching of the DE (θ_{de}) and GE (θ_{ge}) to avoid unnecessary idling losses is an integer decision variable and cannot be handled within the convex optimization algorithm. Thereto, an extra level (c) to handle the optimal ON/OFF switching is introduced.

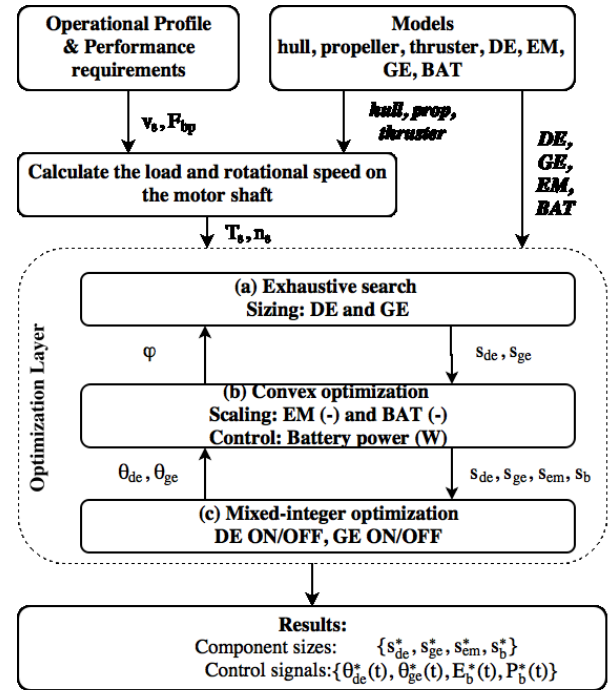


Fig. 3: Proposed coordination and optimization scheme.

A. Optimization criterion

The operational cost ϕ_{op} (€) for a given operational profile are influenced by the fuel power P_f and the power for recharging the battery P_g from the grid (assumption: if the vessel is at the shore, then it will be fully recharged). These are converted to an equivalent cost in euros (€) using the energy prices c_f (€/l) for Diesel and c_e (€/J) for shore-side electricity. The component costs ϕ_{co} (€) are determined by the component sizes and c_{em} (€/kW) and c_b (€/kWh), which are calculated as the depreciation of a component over the

operational profile. The operational profile is measured for one day, hence the component cost depreciations need to be calculated with the ratio of the operational hours per day d_d (hours/day) and the product of the average operational hours per year $\bar{d}_y$ (hours/year), the battery y_b and electric machines lifetime y_{em} in (years), respectively. The total cost or minimizing criterion ϕ (€) is given by,

$$\phi = \phi_{op} + \phi_{co} = \int_{t_0}^{t_f} L(t) dt \quad (9)$$

with

$$L(t) = \underbrace{[\mu_f \quad \mu_{kWh}] \begin{bmatrix} P_f(t) \\ P_g(t) \end{bmatrix}}_{\text{operational cost}} + \underbrace{\begin{bmatrix} \frac{c_b}{y_b} & \frac{c_{em}}{y_{em}} \end{bmatrix} \begin{bmatrix} s_b \\ s_{em} \end{bmatrix} \frac{d_d}{\bar{d}_y}}_{\text{component cost}} \quad (10)$$

and $\mu_f = \rho_f c_f / h_f$ (€/J) and $\mu_{kWh} = c_e / 3.6 \cdot 10^6$ (€/kWh) with the fuel density ρ_f (l/g) and the lower heating value for Diesel h_l (J/g). The minimization problem, as function of the defined controlled states $x_c(t)$, inputs $u_c(t)$ and design parameters x_p is given by Eq. (11) in Table I.

B. Convex component modelling and constraints

For the convex optimization problem, the components are modelled as convex functions. For the DE, the fuel power $P_{f,de}$ can be modelled as a quadratic fit with the torque T_{de} [7], as shown with Eq. (12). The fit coefficients $a_i(\cdot)$ are influenced by the discrete DE size and $n_s(t)$. In a similar fashion, the fuel power consumed by the GE $P_{f,ge}$ can be modelled as a quadratic fit with the power output of the GE P_{ge} , Cf. Eq. (13) with the fit coefficients $b_i(\cdot)$ influenced by the discrete GE size. For the EM, the losses $P_{em,loss}$ can also be modelled as a quadratic fit of T_{em} . To vary the size of the EM, the minimum and maximum torque are scaled by a scaling factor s_{em} [15]. Both the EM losses $P_{em,loss}$ and machine costs c_{em} are assumed to linearly scale with s_{em} . With respect to a baseline EM, the losses of the scaled EM are given by Eq. (14) [9] with the fit coefficients $c_i(\cdot)$ influenced by the size of the base EM and n_s . The non-linear model is convex for $s_{em} > 0$. The BAT pack output power P_b (W) and energy capacity E_b (J) is scaled by multiplying the maximum and minimum usable power and energy capacity per battery cell with the scaling factor s_b (-) equal to the number of battery cells as shown by Eqs. (15) and (16) with the open circuit voltage, V_{oc} (V/cell), the maximum and minimum state-of-charge, ξ_i (-) for $i \in \{\min, \max\}$, the capacity of one battery cell, Q_{cell} (Ah/cell), and the minimum and maximum current delivered by one battery cell, I_{min} and I_{max} (A/cell). Note that the open-circuit voltage V_{oc} is assumed to constant; and, for simplification, the battery and switchboard losses are not incorporated. The remaining constraints are related to the power (17) and torque balance (18); machine (19) and engine output torque (20); GE set output power (21); and, scaling parameters, x_p , (22), (23). Finally, an overview of the defined convex optimization problem and constraints are shown in Table I.

TABLE I: Convex problem for a hybrid tug

$x = \left\{ \underbrace{\theta_{de}(t), \theta_{ge}(t), E_b(t)}_{x_c(t)}, \underbrace{P_b(t)}_{u_c(t)}, \underbrace{s_{em}, s_b}_{x_p} \right\}$	
$\min_x \phi(x_c(t), u_c(t), x_p \mid \Lambda(t)) \quad (11)$	
$P_{f,de}(t) = \theta_{de}(t) \cdot \sum_{i=0}^{i=2} a_i(n_s(t)) \cdot T_{de}^i(t) \quad (12)$	
$P_{f,ge}(t) = \theta_{ge}(t) \cdot \sum_{i=0}^{i=2} b_i \cdot P_{ge}^i(t) \quad (13)$	
$P_{em,loss}(t) = c_2(n_s(t)) \cdot \frac{T_{em}^2(n_s(t))}{s_{em}} + c_1(n_s(t)) \cdot T_{em}(n_s(t)) + c_0(n_s(t)) \cdot s_{em} \quad (14)$	
$P_b(t) \in [I_{min}, I_{max}] \cdot V_{oc} \cdot s_b \quad (15)$	
$E_b(t) \in [\xi_{min}, \xi_{max}] \cdot Q_{cell} \cdot V_{oc} \cdot s_b \quad (16)$	
$P_{em,loss}(t) + P_{em}(t) + P_{aux}(t) \leq P_b(t) + \theta_{ge}(t) \cdot P_{ge}(t) \quad (17)$	
$T_{em}(t) + \theta_{de}(t) \cdot T_{de}(t) \geq T_s(t) \quad (18)$	
$T_{em}(t) \in [T_{em,min}(n_s(t)), T_{em,max}(n_s(t))] \cdot s_{em} \quad (19)$	
$T_{de}(t) \in [T_{de,min}(n_s(t)), T_{de,max}(n_s(t))] \quad (20)$	
$P_{ge}(t) \in [P_{ge,min}, P_{ge,max}] \subset \mathbb{R} \quad (21)$	
$s_{em} \in [s_{em,min}, s_{em,max}] \subset \mathbb{R} \quad (22)$	
$s_b \in [s_{b,min}, s_{b,max}] \subset \mathbb{R} \quad (23)$	
$\forall t \in [t_0, t_f] \subset \mathbb{R}^+$	

C. Mixed-integer optimization: engine ON/OFF switching

For determining the discrete DE and GE ON/OFF states, different methods, e.g., based on dynamic programming (DP) [16] or heuristics [7] can be applied. An alternative method is proposed here using convex and mixed-integer (MI) optimization based on the combining techniques from [17] and [16]. The level ‘(c) ON/OFF switching level’ consists of two parts, which are shown in Fig. 4 from top to down (Cf. also Fig. 3). First, the convex optimization takes place at ‘(b) Convex optimization’. Thereby, the DE is assumed to be ON for all feasible $n_s(t)$ and the GE is ON for $T_s(t) > 0$, $n_s(t) > 0$ or $P_{aux}(t) > 0$ and, therefore, defines the initial set $\{\theta_{ge}^0(t), \theta_{de}^0(t)\}$. These conditions give all the feasible ON times for the DE and GE before the iteration starts. Therefore, for the subsequent iteration steps, holds,

$$\theta_{de}^j(t) \subseteq \theta_{de}^0(t) ; \theta_{ge}^j(t) \subseteq \theta_{ge}^0(t) \quad (24)$$

with j the number of iterations. From this first convex optimization, the sizing parameters s_{em}^0, s_b^0 are taken to level (c). Within the level (c), first a set of equivalent cost problems, denoted as H_c^j , is optimized similar to the optimization problem in Table I for four different ON/OFF switching cases, denoted as c : $c = 1$: $\{\theta_{de}, \theta_{ge}\} = \{1, 1\}$; $c = 2$: $\{\theta_{de}, \theta_{ge}\} = \{1, 0\}$; $c = 3$: $\{\theta_{de}, \theta_{ge}\} = \{0, 1\}$; and, $c = 4$: $\{\theta_{de}, \theta_{ge}\} = \{0, 0\}$, using:

$$H_c^j = L_c^j(t) + \lambda^j(t) \cdot u_c(t) \quad (25)$$

with $\lambda^j(t)$ denoting the equivalence factor between fuel and battery energy consumption. For each subsequent iteration step

j , the equivalent cost value of every case c at every time instant $H_c^j(t)$ is the input for the ‘mixed-integer minimization problem’:

$$\min_{u_c^j, \theta_c^j} \phi_{mi}^j = \min_{u_c^j, \theta_c^j} \int_{t_0}^{t_f} \sum_{c=1}^4 H_c^j(t) \cdot \theta_c^j(t) dt \quad (26)$$

with $\theta_c^j(t) \in \{0, 1\}$ as the new defined binary decision variable for a case being inactive or active, respectively. Moreover, the modified control problem is additionally constrained by,

$$\sum_{c=1}^4 \theta_c^j(t) = 1 \quad (27)$$

$$\int_{t_0}^{t_f} \sum_{c=1}^4 P_{b,c}^j(t) \cdot \theta_c^j(t) dt \in [\xi_{min}, \xi_{max}] Q_{cell} V_{oc} s_b^j, \quad (28)$$

with $P_{b,c}^j(t)$ the battery power for every case at every time instance at a certain iteration step. The resulting $\theta_{ge}^j(t)$ and $\theta_{de}^j(t)$ are an input for the second convex optimization (after the initialization) in level (b). Notice that a change in $\theta_{de}^j(t)$ and $\theta_{ge}^j(t)$ is strongly influencing s_{em}^j and s_b^j and vice versa. In addition, the equivalence factor as an output of the convex optimization λ_{out}^j and used as input λ_{in}^j in the MI optimization influences the optimized control signals. In similar manner as in [16], the λ_{in}^j is altered with an ‘update law’ assuring that the optimized parameters and variables meet a convergence criteria. If the absolute change in total cost, the battery and electric machine size, are all lower than a predefined tolerance value, i.e., $e_{\phi, tol} = 5 \cdot 10^{-4}$ (€), $e_{em, tol} = 10^{-2}$ (-) and $e_{b, tol} = 5$ (cells), then the iteration process is terminated and the optimization results, denoted with * are returned.

D. Design optimization results

The iterative optimization for the convex and mixed-integer problems is performed using solvers from the licensed MOSEK tool [18] for the set of constraints and inputs, as listed in Table II. The simulation results are listed in Table III. The complete optimization was executed in 482 seconds² including ten iterations for ϕ^j , s_{em}^j and s_b^j required to converge.

The corresponding operational costs ϕ_{op} (€) in terms of the electricity and fuel cost and the component costs ϕ_{co} (€) for the EM and BAT at each iteration step are plotted in Fig. 5 and Fig. 6, respectively.

IV. CONCLUSION

The combined optimization of the hybrid power train and the control design for a tug has been discussed. The multi-objective and non-linear character of the problem requires, usually, a large computation time and a global optimal results can not be guaranteed. In order to speed up the calculation time with acceptable modelling error on the main components, the efficiency models are transformed into convex models. The convex iterative approach converges towards an optimum in a short amount of time (8 minutes). In this work, the convex

²CPU: i7, 720 QM, 1.60 Ghz, 4 GB RAM

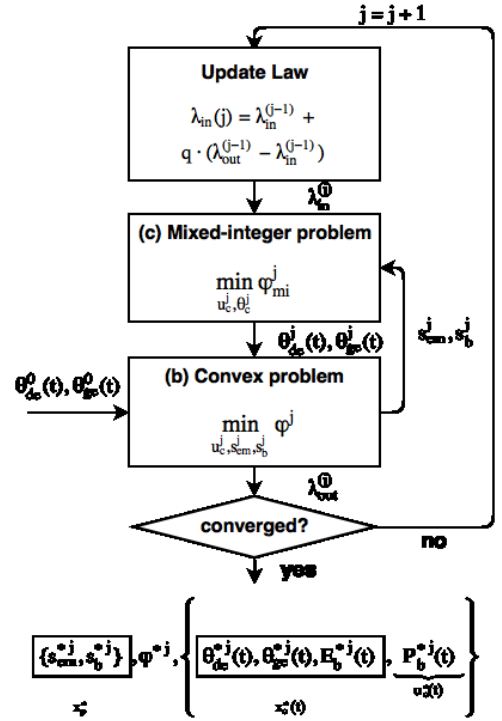


Fig. 4: Iteration scheme of level (b) and (c) of the optimization layer for given values s_{de} and s_{ge} from level (a).

TABLE II: Model input parameters

Symbol	Value	SI unit
$\{T_{de, min}, T_{de, max}\}$	$\{0, 14\}$	(kNm)
$\{T_{em, min}, T_{em, max}\}$	$\{-3, 3\}$	(kNm)
$\{s_{em, min}, s_{em, max}\}$	$\{0.5, 1.4\}$	(-)
$\{s_{b, min}, s_{b, max}\}$	$\{500, 2000\}$	(-)
s_{de}, s_{ge}	1	(-)
$c_e / 3.6 \cdot 10^6$	0.15	(€/kWh)
c_f	1.1	(€/l)
c_{em}	33.3	(€/kW)
c_b	1389	(€/kWh)
d_d	6h40'	(hour minutes/day)
$\bar{d}_y$	2433	(hour/year)
y_b	15	(year)
y_{em}	20	(year)
q	0.2	(-)

TABLE III: Minimized costs per day, ϕ^* at the optimal design $\{s_{em}^*, s_b^*\} = \{1.5, 640\}$

Component cost function	Cost (€)
Fuel cost DE	179
Fuel cost GE	62
Grid charging cost	10
Component cost BAT	23
Component cost EM	8
Total cost, ϕ^*	283

optimization is integrated with a mixed-integer optimization scheme for determining the optimal discrete control inputs to the problem and design parameters. To extend the model extra

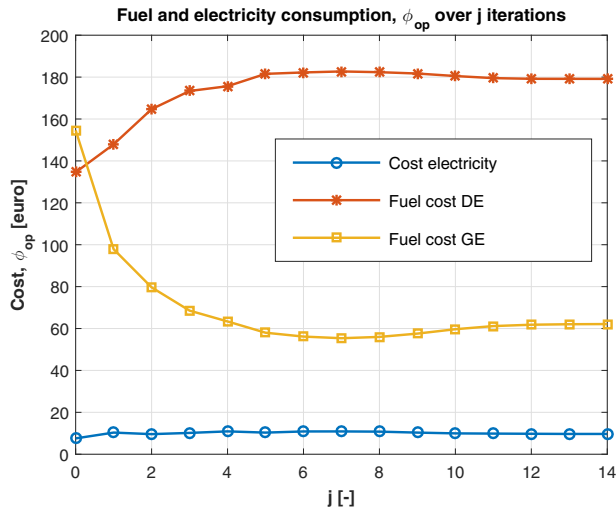


Fig. 5: The operational costs at each iteration step.

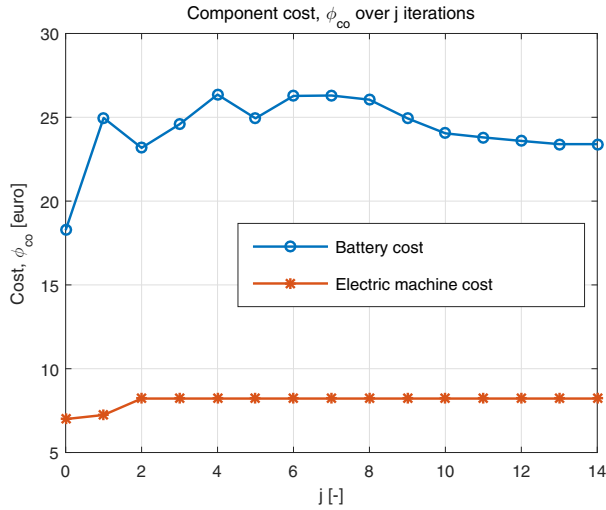


Fig. 6: The component costs at each iteration step.

states can be added to the model without much increase in computational time.

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